

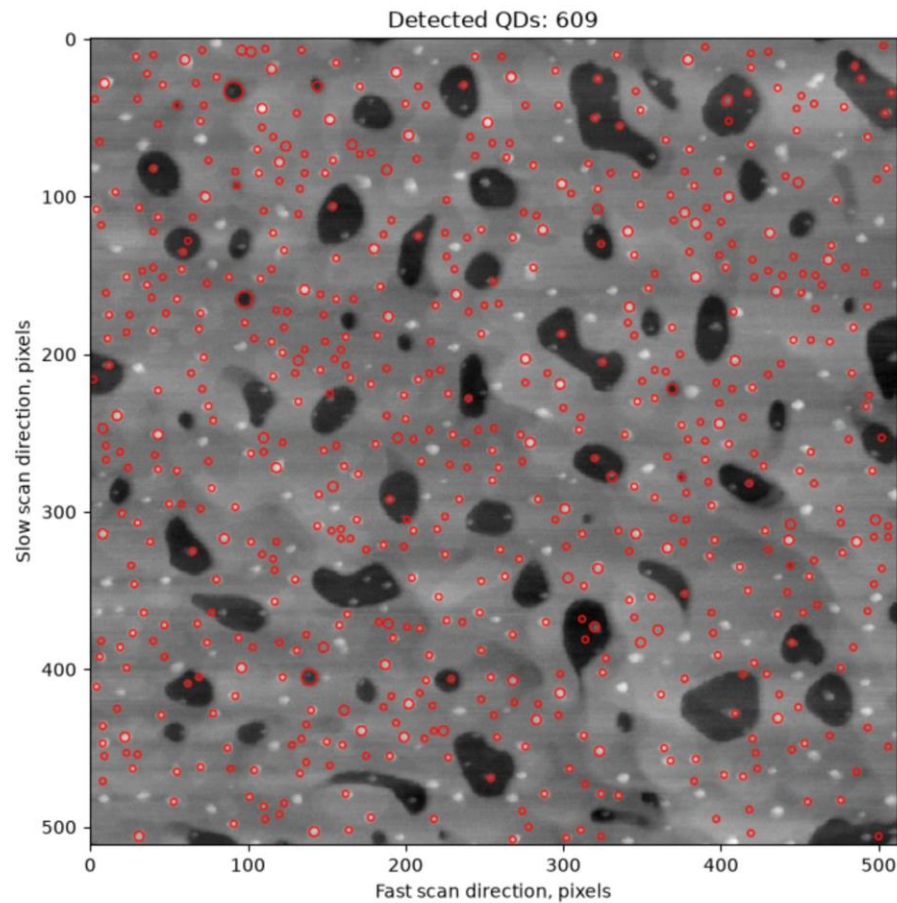
Week 5 Progress Meeting presentation

By Poline S

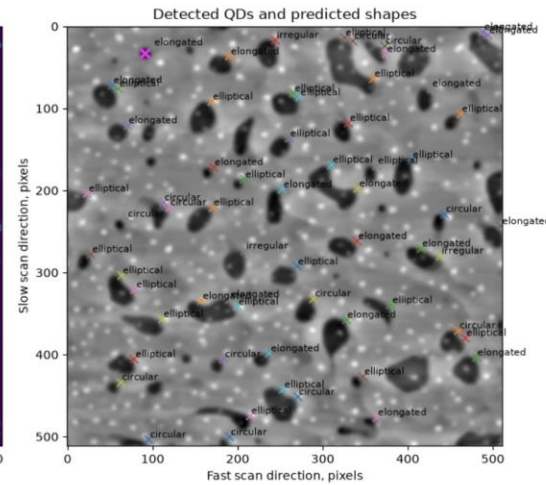
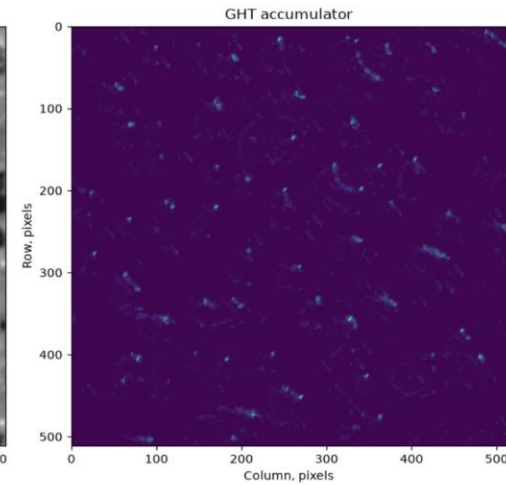
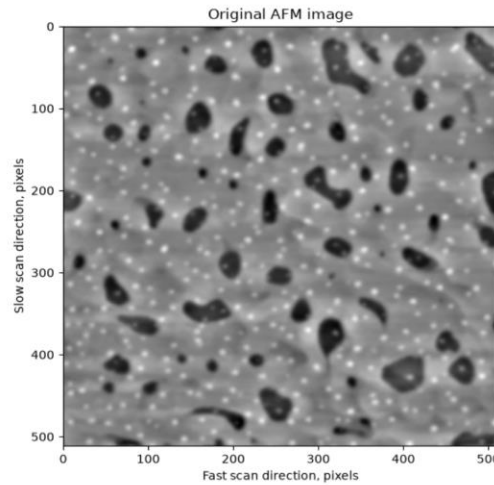
Summary

- Model now trained on more scans (2 scans labelled by Chenuye, 2 1nm, 2 2nm, 2 3nm)
- Pixelwise correlation, recall, precision, f1 etc calculated for each of the models and overlay of their predicted binary scans with the ground truth was created
- Tried HT, Watershed and Geometric contours for QD detection -> solved my issue with circularity, **which is now being replaced with eccentricity**
- Testing app modified to include information about the dot by hovering cursor on it for it to be easier that the dot fullfills the height requirement
- Dots within GaN, and outside GaN - changed to exclude dots on the edges
- Reorganised the code to include functions for QD detection and error analysis in separate files
- Moved around Data Conversion and Ground Truth

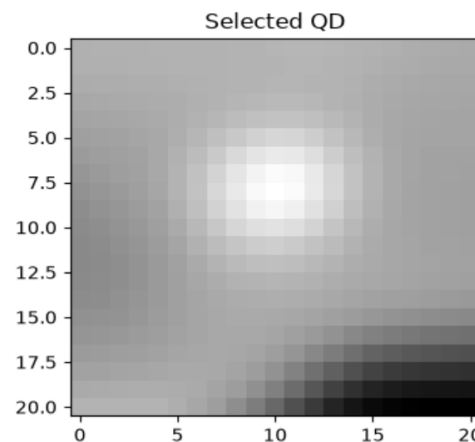
Hough Transform



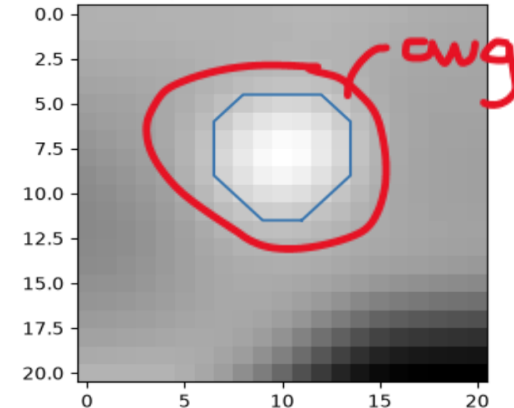
Number above 0.75: 572
Number above 0.745: 609



Didn't end up performing too well, but was very useful for finding out what was the issue with calculating the circularity of the QD, **eccentricity + symmetry/ antisymmetry.**



circular
Circularity = 1.055, axis ratio = 0.991

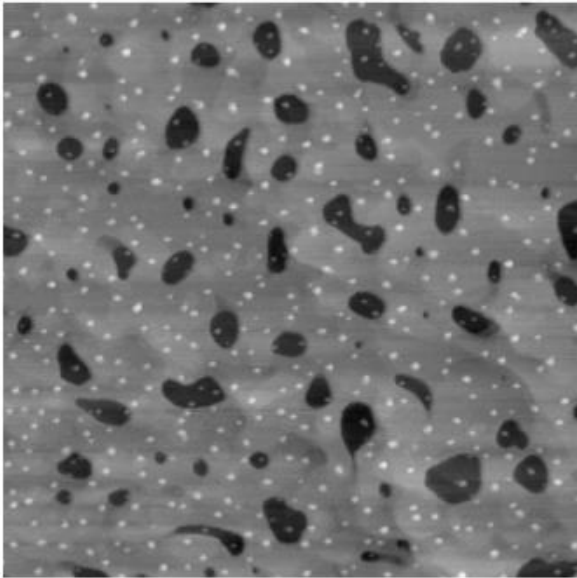


Dilate the QD by a factor of the avg height of the back QD AND detection

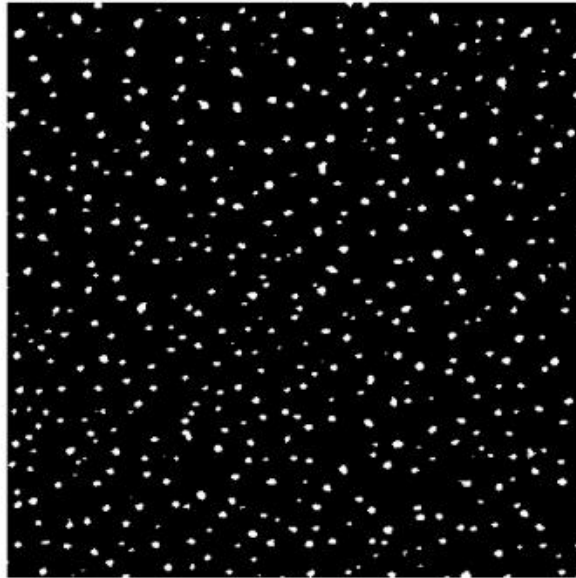
Watershed

Detected QDs: 472

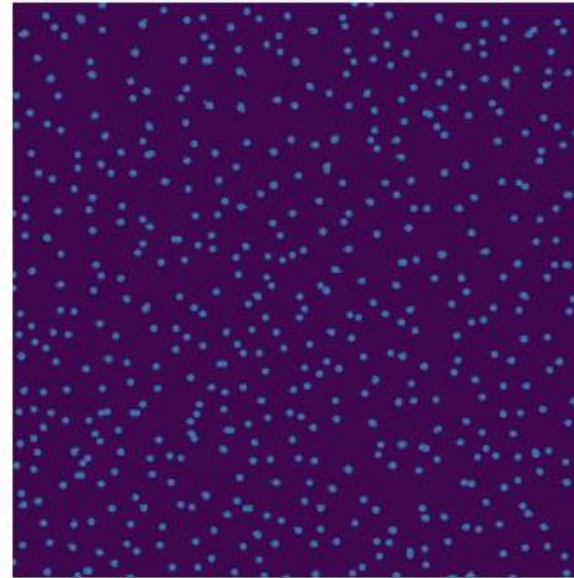
Preprocessed AFM



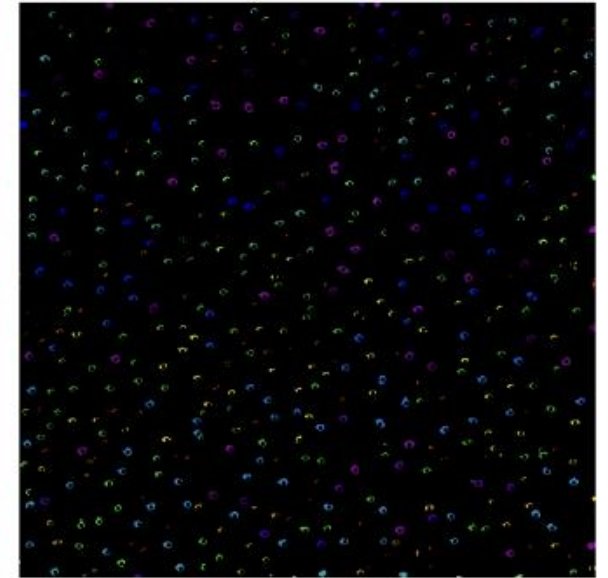
QD candidate mask



Distance transform + markers

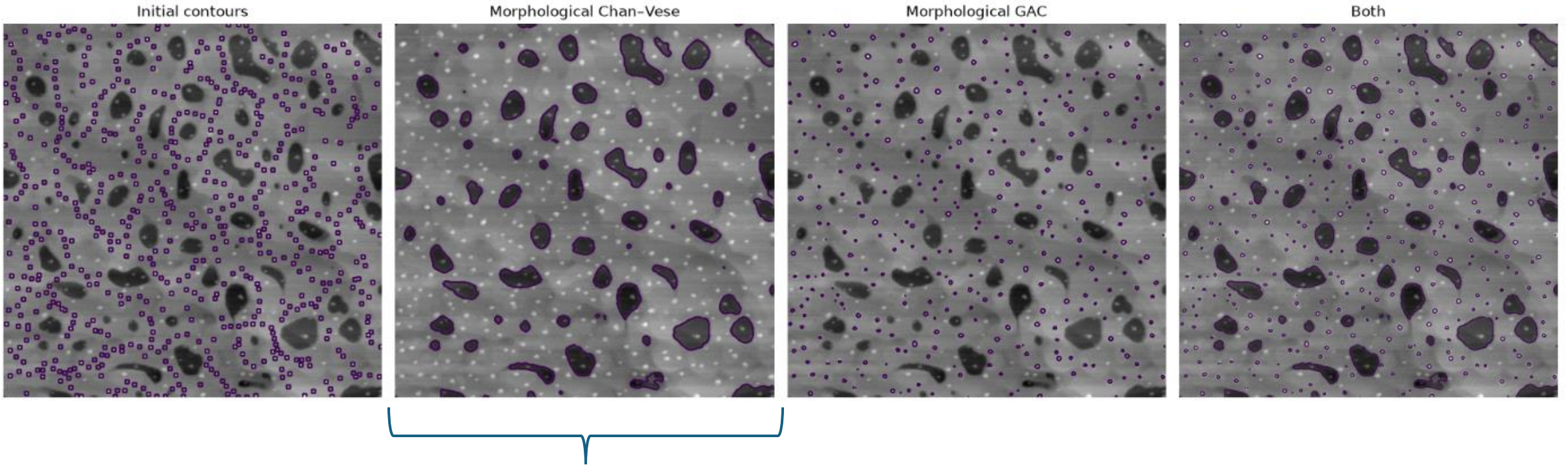


Watershed QDs: 472



```
watershed_labels = watershed(-distance, markers, mask=qd_mask)
```


Contour Detection



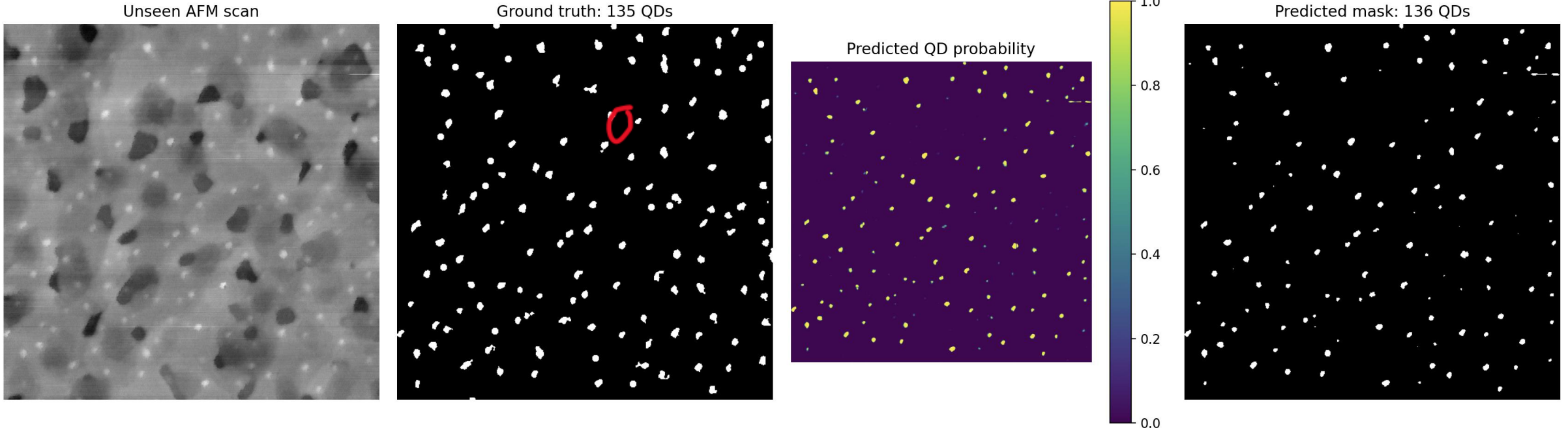
Could be useful instead of having to choose thresholding method for GaN pit detection.

*These are new, so didn't yet include them into error analysis

CNN for QD detection

Rotate scan3 to have more tests

Height -> make the model predict height



*I may have labelled some extra dots on other scans which is why the model is more generous..

- 2 layer CNN, uses sigmoid as an activation function
- Low confidence, so may need to label more scans (currently 8 scans but all augmented + 1 test scan)

-> Pixelwise correlation: 0.62
-> Dice coefficient: 0.58
-> IoU: 0.41
-> Ground truth vs probability-map correlation: 0.62

Errors calculations

	Algorithm	Stage	Predicted count	True count	Count error	Absolute count error	Pixel precision	Pixel recall	Pixel F1	Dice	...	Localisation mean (px)	Localisation median (px)	Localisation RMSE (px)	Localisation maximum (px)
0	LoG	Before height filter	232	135	97	97	0.2037	0.7590	0.3212	0.3212	...	1.8139	1.5660	2.1762	5.0000
1	LoG	After height filter	138	135	3	3	0.3761	0.6897	0.4867	0.4867	...	1.6893	1.4250	2.0341	5.0000
2	CAD	Before height filter	253	135	118	118	0.1990	0.7435	0.3139	0.3139	...	1.8254	1.5834	2.1839	5.0000
3	CAD	After height filter	141	135	6	6	0.3803	0.6711	0.4855	0.4855	...	1.6938	1.4120	2.0403	5.0000
4	Otsu	Before height filter	322	135	187	187	0.1737	0.7528	0.2823	0.2823	...	1.8057	1.5660	2.1647	5.0000
5	Otsu	After height filter	154	135	19	19	0.3707	0.6722	0.4779	0.4779	...	1.7138	1.4250	2.0468	5.0000
6	HT	Before height filter	491	135	356	356	0.1317	0.0711	0.0924	0.0924	...	1.8747	1.8679	2.1480	4.3791
7	HT	After height filter	28	135	-107	107	0.9026	0.0662	0.1233	0.1233	...	1.7720	1.5965	2.0484	4.3791
8	PMAD	Before height filter	117	135	-18	18	0.1466	0.5985	0.2355	0.2355	...	1.8272	1.7096	2.0613	4.3904
9	PMAD	After height filter	73	135	-62	62	0.2025	0.5253	0.2923	0.2923	...	1.8568	1.7096	2.0847	4.3904
10	PC	Before height filter	295	135	160	160	0.2644	0.0073	0.0143	0.0143	...	1.5741	1.3881	1.8239	5.0000
11	PC	After height filter	101	135	-34	34	0.7723	0.0073	0.0145	0.0145	...	1.5490	1.3848	1.7982	5.0000
12	SIFT	Before height filter	102	135	-33	33	0.5585	0.3384	0.4215	0.4215	...	1.4719	1.3829	1.7072	4.0963
13	SIFT	After height filter	72	135	-63	63	0.7545	0.3313	0.4604	0.4604	...	1.4914	1.3848	1.7336	4.0963
14	WS	Before height filter	2244	135	2109	2109	0.2385	0.8558	0.3731	0.3731	...	1.5299	1.3332	1.8434	5.0000
15	WS	After height filter	213	135	78	78	0.5744	0.8150	0.6739	0.6739	...	1.5958	1.3881	1.9070	5.0000
16	GW	Before height filter	931	135	796	796	0.1794	0.0157	0.0289	0.0289	...	1.7425	1.5369	2.0286	5.0000
17	GW	After height filter	240	135	105	105	0.6208	0.0140	0.0274	0.0274	...	1.7153	1.4899	2.0002	5.0000

Need to add error calculations for height, pixelwise area and radius of the QD, given Xin's data

Watershed > LoG $\approx$ Chambolle Anisotropic Diffusion $\approx$ Otsu > Scale Invariant Feature Transform > PMAD > HT > GW > Phase Congruency

1. Number of false positives and false negatives - > how far a dot should be from the GT to say its an FT
2. Add more tests for QD finding in the algorithm
3. Prioritise QD count and match between GT and predicted QD centres -> threshold?

-> **Best segmentation:** Watershed

-> **Best count:** LoG

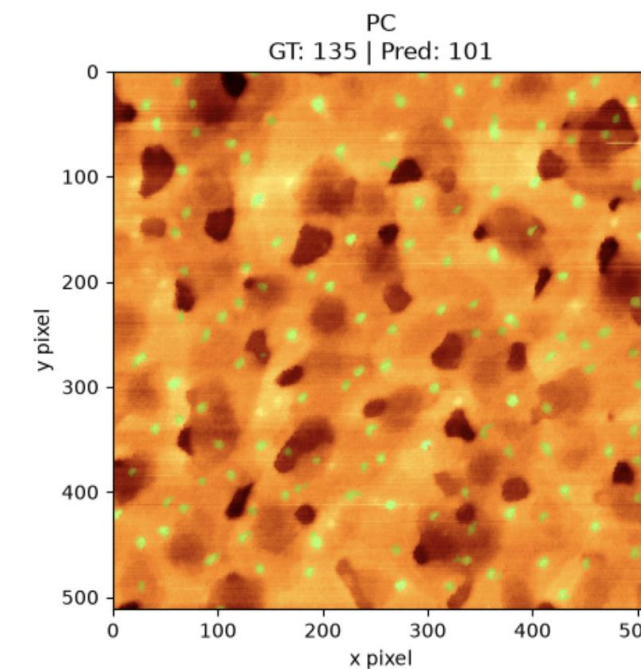
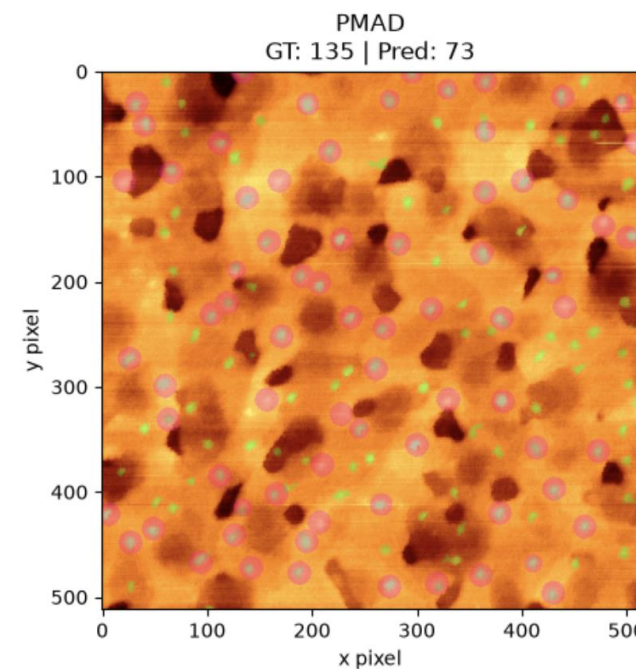
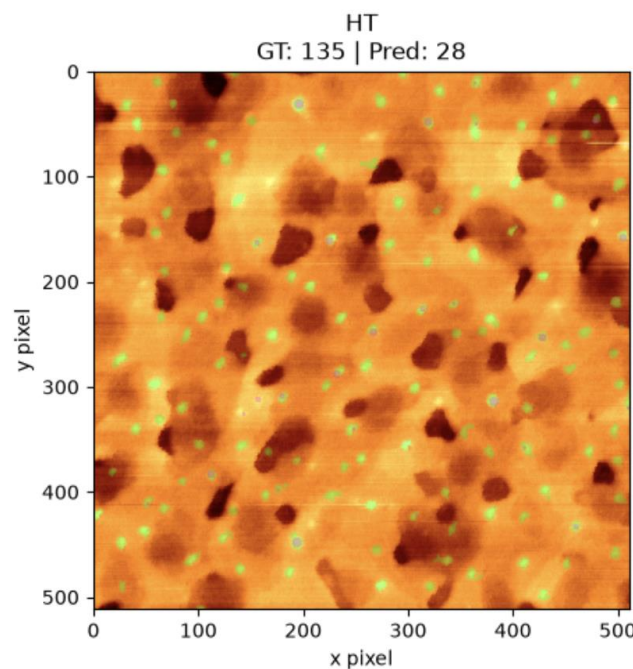
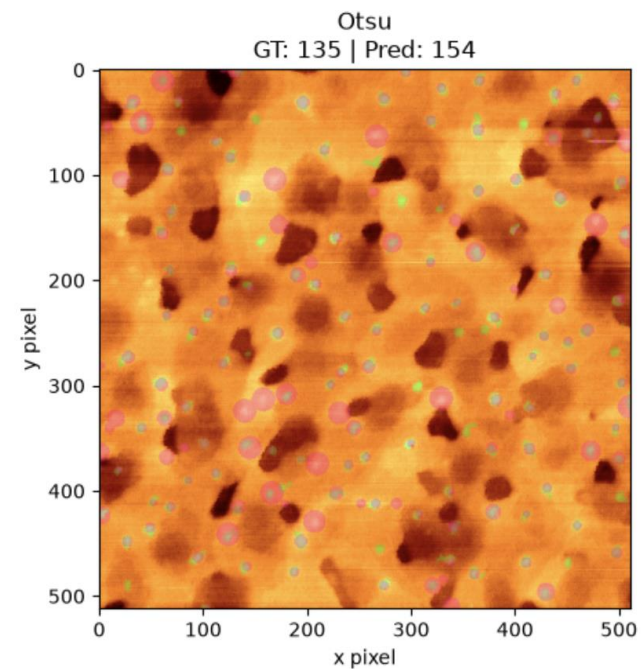
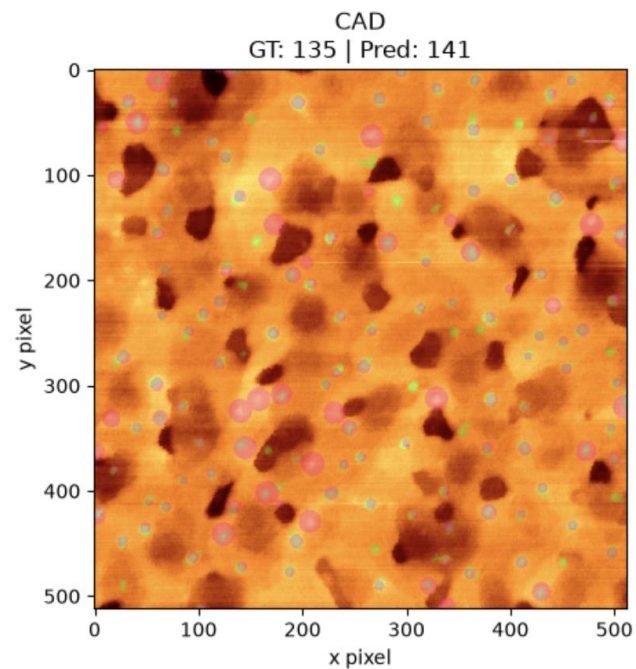
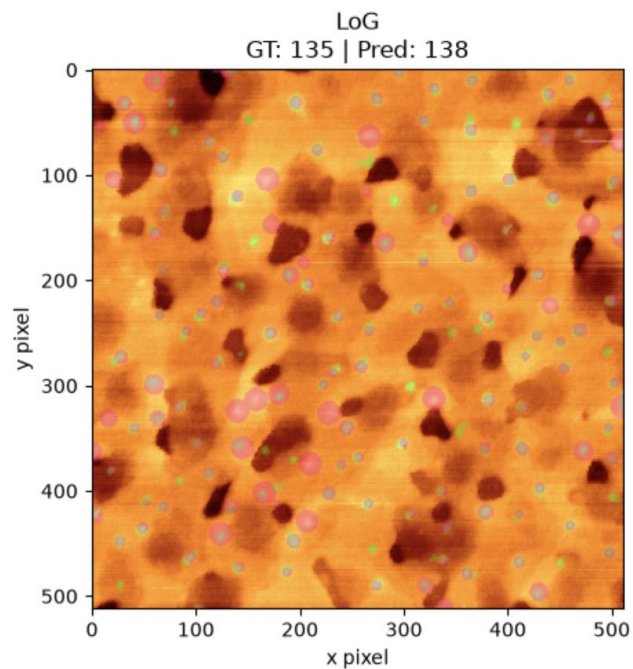
-> **Best conservative/high-precision detector:** SIFT (HT has higher precision, but its recall is catastrophically low)

-> **Worst:** PC, followed by GW

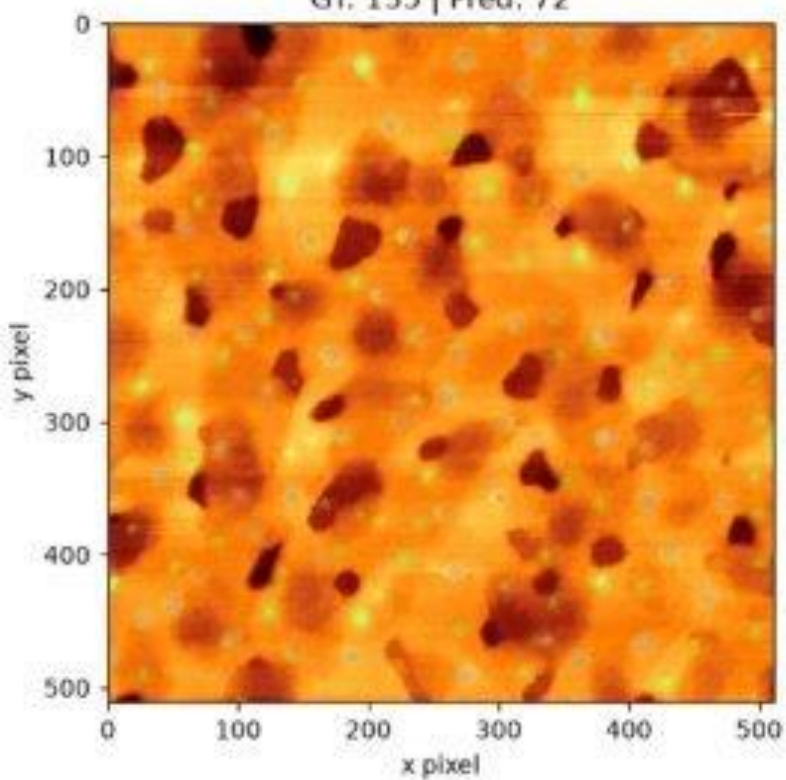
Overlay

Use different
shape for
labelling

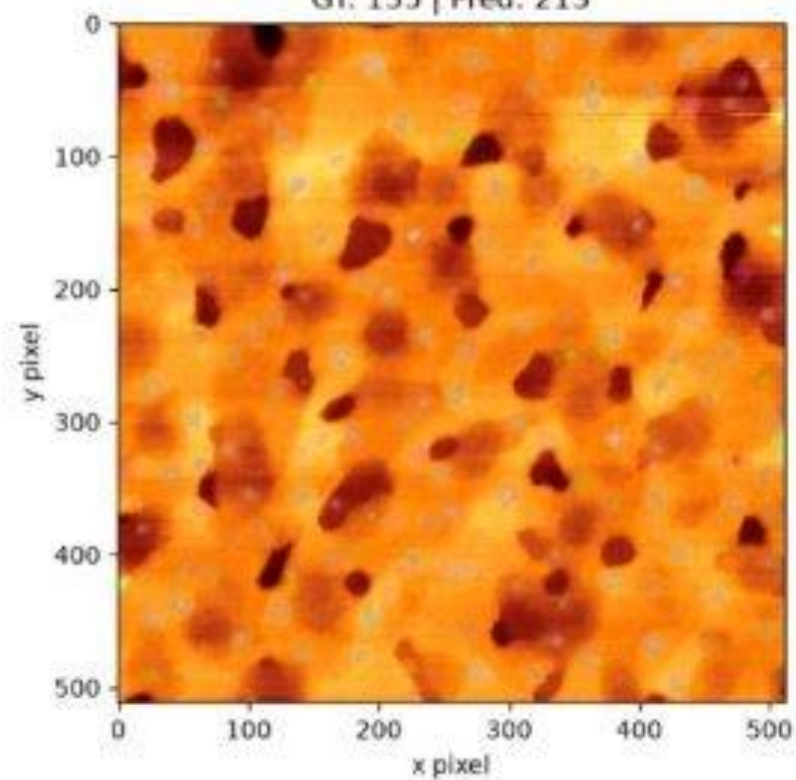
Filled detection overlays — After Height Filter
Green = Ground truth, Magenta = Prediction



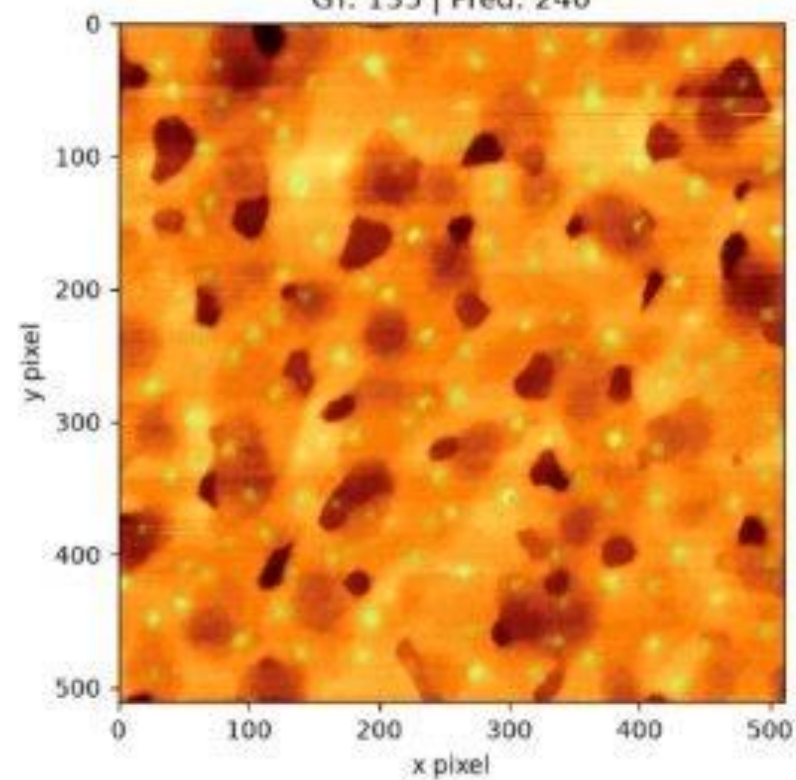
SIFT
GT: 135 | Pred: 72



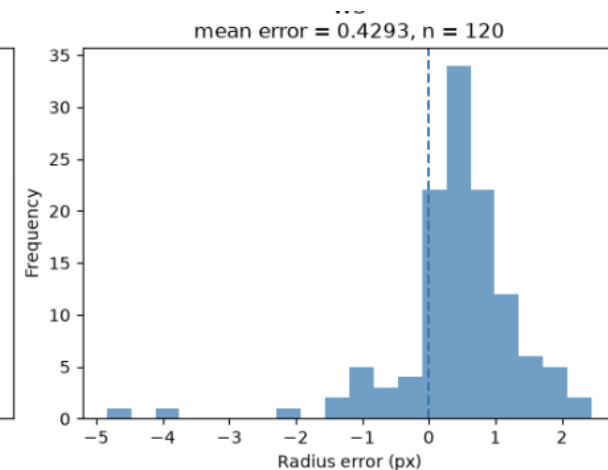
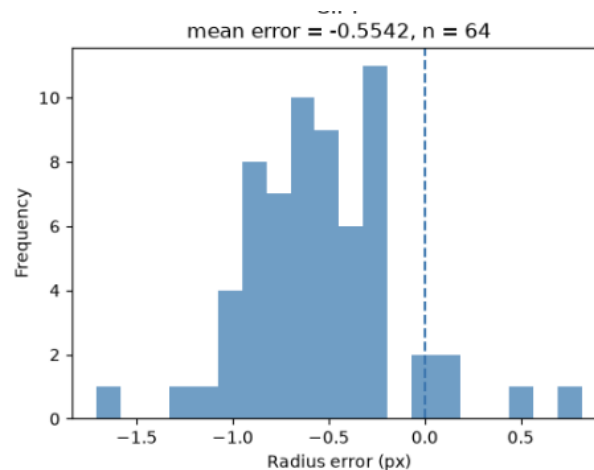
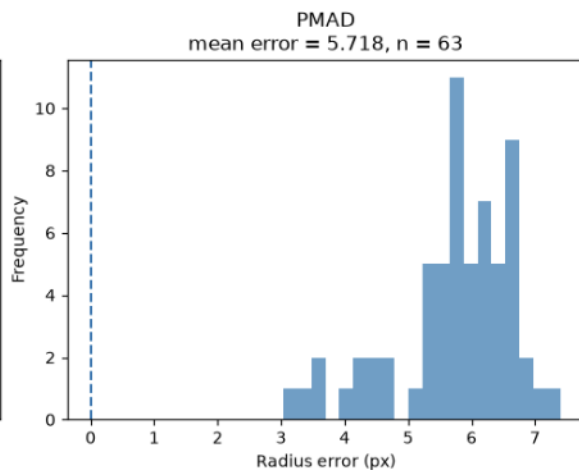
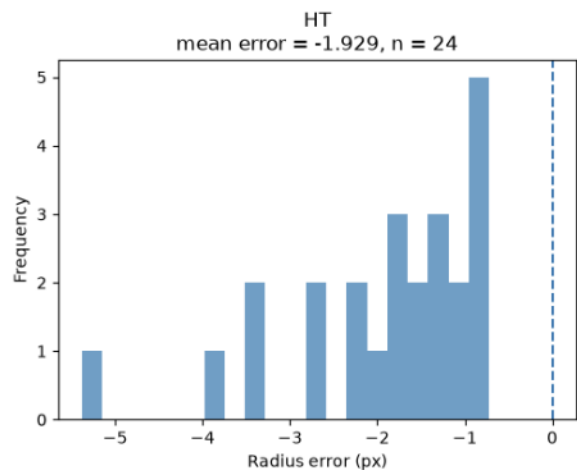
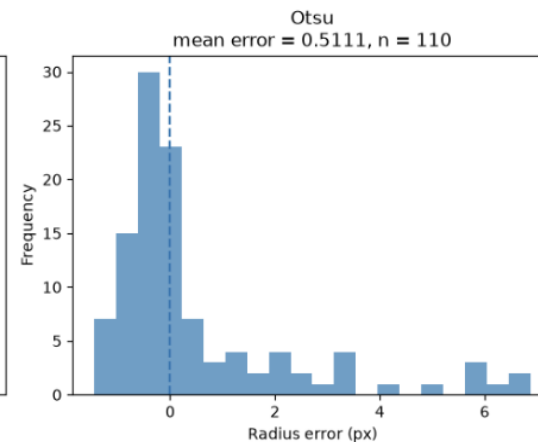
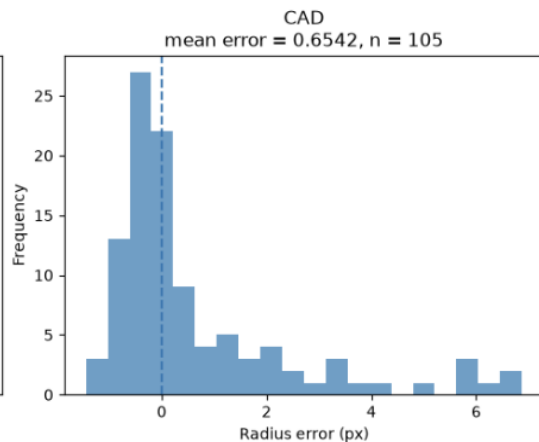
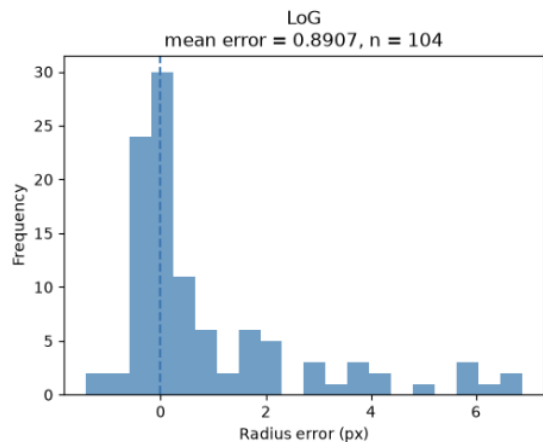
WS
GT: 135 | Pred: 213



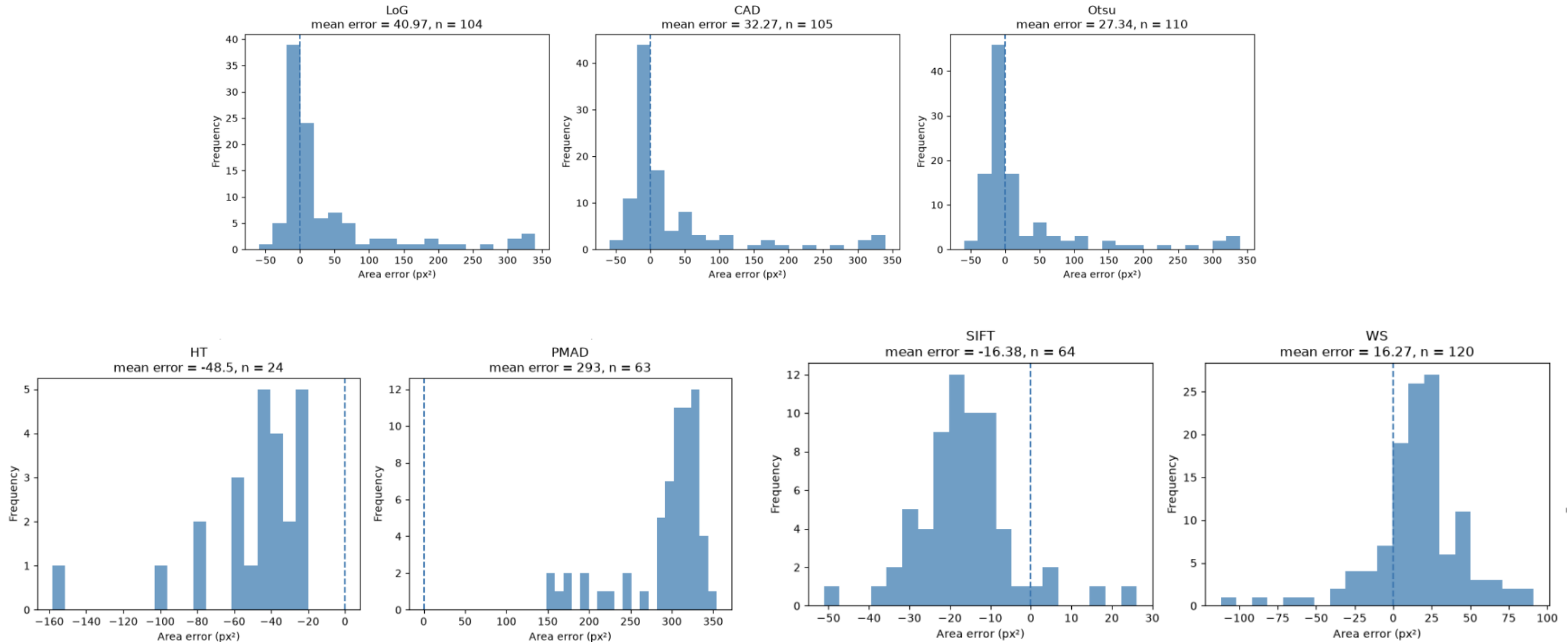
GW
GT: 135 | Pred: 240



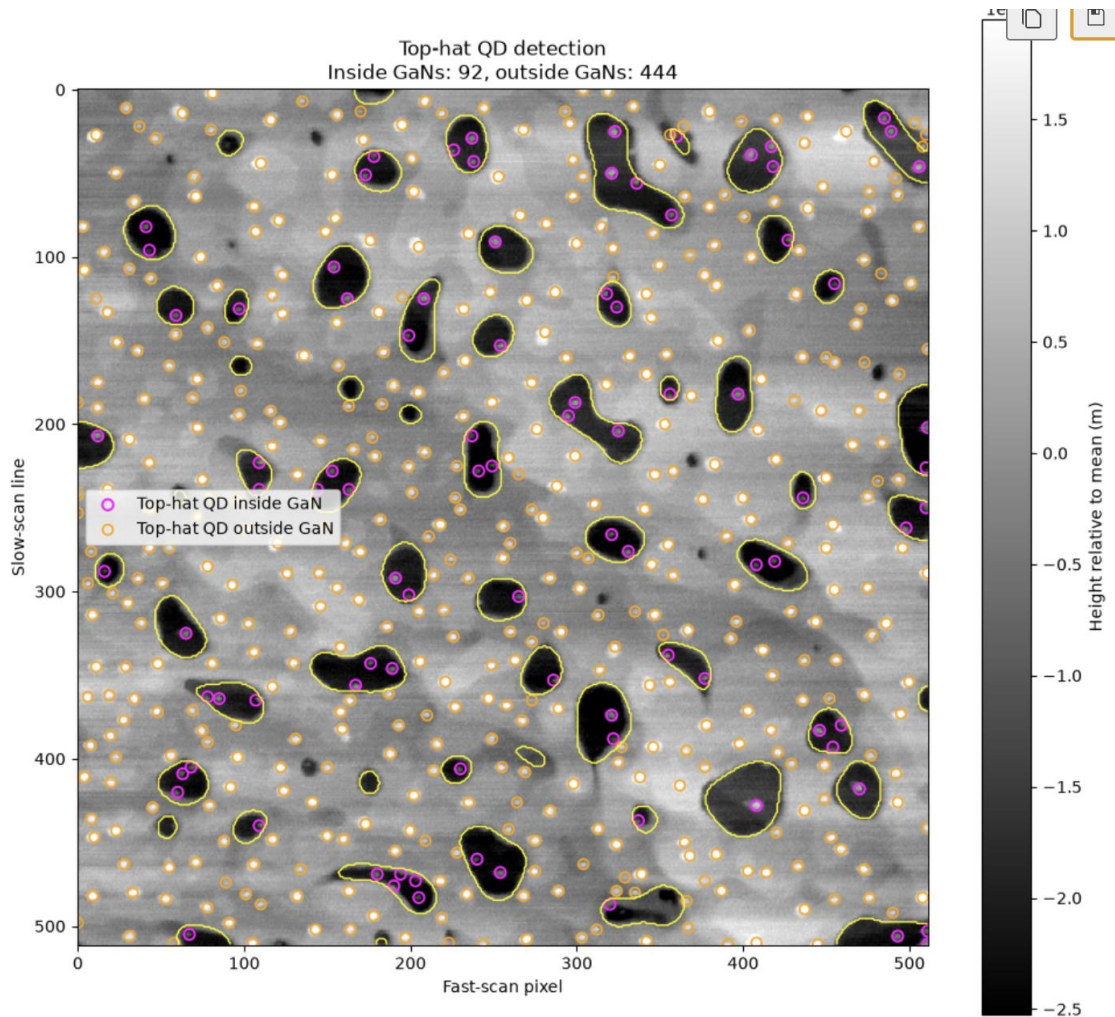
Calculating errors in Radius from GT binary mask



Calculating errors in Area from GT binary mask

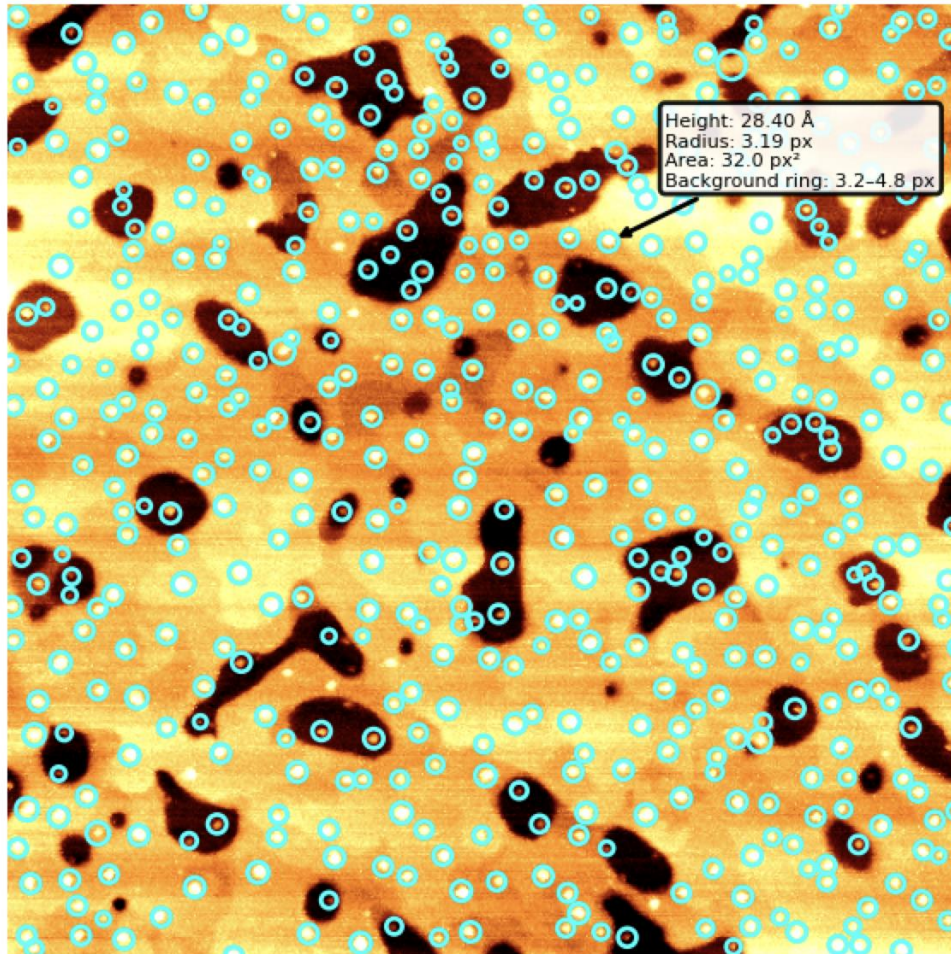


QDs in GaN layer detection improved



Labelling App improvements

AFM image | QDs ≥ 6 Å: 528



Final binary mask

